



Fatemeh Aghaei

*MSc Bioinformatics (Distinction) | NGS, multi-omics & biomarker discovery |
Reproducible analysis in R & Python*

Profile

Bioinformatics MSc graduate (Distinction) with hands-on experience in high-throughput DNA sequence analysis, reproducible R and Python workflows, and turning technical analyses into tools that non-specialists can run themselves. Four years in a GMP laboratory building automated data pipelines that connect people, data and systems. Comfortable on the Linux command line with Git version control, Galaxy and FAIR-aligned, reproducible analysis. My work spans NGS and RNA-seq analysis, multi-omics integration, and biomarker discovery, with a focus on turning complex biological data into clear, reproducible results.

Education and training

2023–2025 **MSc Bioinformatics**, *University of Birmingham*, Birmingham, United Kingdom, Grade: Distinction | Graduated Dec 2025.

- Hands-on high-throughput sequence analysis: NGS and RNA-seq data processing, quality control and downstream analysis using R and Python in reproducible, documented workflows.
- Coursework in Applied Genomics & Next-Generation Sequencing, Statistical Machine Learning, metabolomics and multi-omics integration; built and ran analysis pipelines on Galaxy, with initial exposure to SnakeMake / Nextflow.
- Thesis: investigated ARNT as a prognostic biomarker in non-muscle-invasive bladder cancer, integrating transcriptomic, CNV, clinical, survival and pathway analyses.

2017–2021 **BSc Cellular and Molecular Biology – Microbiology**, *Azad University, Tehran Medical Branch*, Tehran, Iran.

- Foundation in microbiology, molecular biology, genetics and evolution, including environmental and applied microbiology and biomedical laboratory science.

Working experience

2022–Present **Senior Microbiology Technician**, *Johnson & Johnson Innovative Medicine, QC-Microbiology*, Leiden, the Netherlands.

- Design and build automated data pipelines and reporting tools so non-technical lab staff can retrieve, visualise and report QC data themselves; initiated a Shiny-based QC reporting concept that scaled into a global Johnson & Johnson initiative.
- Work with structured data platforms (Databricks), API / database connections and reproducible reporting (R, Quarto) to improve data accessibility, consistency and GMP compliance.
- As a QC Digital & Technology Council representative, translate laboratory workflow needs into digital, automation-driven solutions across teams – bridging technical and non-technical colleagues.
- Support GMP microbiology and CAR-T manufacturing through testing, data review, trend analysis and investigation support.

2026–Present **Volunteer Data Analyst**, *Statistics Without Borders*, Remote.

- Clean, curate and validate large structured healthcare datasets in R, producing reproducible, well-documented analysis and visualisation for a child-survival and hospitalisation-risk study.

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Portfolio

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- 2019 **Microbiology Laboratory Technician**, *Naft Hospital*, Tehran, Iran.
- Performed routine microbiological testing in a clinical laboratory, supporting accurate result documentation and reporting.

Selected projects

- 2026 **16S rRNA amplicon analysis (DADA2, R)**, *Self-directed project*, GitHub.
- Implemented a 16S rRNA amplicon sequencing workflow in R with DADA2 – quality filtering, error learning, denoising, ASV inference, chimera removal and taxonomic assignment – turning raw marker-gene reads into a taxonomically annotated ASV table.
- 2025 **Multi-omics analysis of pollutant stress in *Daphnia magna***, *MSc project – Computational Biology of Complex Systems*, R.
- Integrated transcriptomics (DESeq2) and LC-MS metabolomics (PCA, differential metabolites, KEGG / HMDB pathway enrichment) using RGCCA and DIABLO / mixOmics to characterise dose-dependent chemical stress responses in *Daphnia magna*; presented the work to the cohort.
- 2026 **Automated QC reporting tool (R / Shiny)**, *Johnson & Johnson – Lean Six Sigma Yellow Belt*.
- Designed a Shiny-based reporting tool that replaced manual data gathering with automated data retrieval, visualisation and GxP-ready report generation; the concept scaled into a global initiative across sites, improving reporting efficiency, consistency and compliance.

Technical skills

Programming & version control:	R (advanced), Python, Git / GitHub, Bash; Quarto, LaTeX
Sequence analysis & bioinformatics:	Amplicon (ASV) sequence analysis (DADA2), NGS / RNA-seq analysis, quality control, differential expression, CNV analysis, pathway enrichment, survival & biomarker analysis, multi-omics integration
Pipelines, Linux & reproducibility:	Linux command line, Galaxy (hands-on); familiar with SnakeMake / Nextflow; cloud / remote compute (Google Cloud); reproducible, FAIR-aligned workflows
Data, ML & integration:	data wrangling, statistical modelling, machine learning, visualisation (Shiny, Tableau, ggplot2, PCA); structured data (Databricks), API / database connections

Selected courses & certifications

- 2023–2024 **Applied Genomics and Next-Generation Sequencing**, *University of Birmingham*, part of MSc Bioinformatics, Processing, integration and analysis of large-scale genomic and sequencing data.
- 2023–2024 **Statistical Machine Learning**, *University of Birmingham*, part of MSc Bioinformatics, Machine learning models for data-driven insight, predictive analytics and decision support.

Languages

English (Fluent), Dutch (Intermediate), Persian (Native).

Volunteering

- 2022–Present **Volunteer**, *Stichting NL Cares*, Amsterdam, the Netherlands.